

Fig. 2: An actual-size, single-side PCB for the electronic chanting device

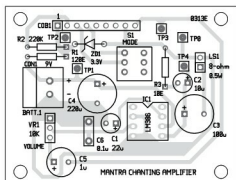


Fig. 3: Component layout for the PCB

The mode to play a particular song is selected through mode-selection switch S1. When mode switch S1 is opened, a *bhajan* plays on. You can toggle the switch to select the next *bhajan*. When S1 is closed, a particular song plays continuously unless power is switched off. LM386 is a low-power audio amplifier. Audio input is fed to its pin 3 via capacitor C5 and volume control VR1. The audio output is available at its pin 5.

Construction and testing

An actual-size, single-side PCB for the electronic chanting amplifier is shown in Fig. 2 and its PCB layout in Fig. 3.

Mount all the components, except COB1, on the PCB. COB1 should be fixed using glue. Use IC base for LM386. This will help you remove the IC easily in case it becomes faulty. There are nine pins available in the

COB and therefore connecting points for the same are provided on the PCB. However, only six pins have been used here. Extend six wires from the PCB and solder each wire to the respective pin on the COB as shown in the schematic diagram. Use a good-quality speaker. We have tested the circuit using an 8-ohm, 0.5W speaker.

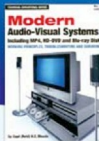
Connect the 9V PP3 battery at CON1 as shown in Fig. 3. You will hear the *Jai Ganesha Jai Ganesha Jai Ganesha Deva* song. If the volume is not loud enough, adjust control knob VR1 to increase the volume. Depending on the position of switch S1, the song will either repeat or stop after it completes. To play the next song, toggle switch S1.

If you don't hear sound from the speaker, first check the power supply. To help you troubleshoot the circuit, various test points are listed in the table. You need a voltmeter to check the voltages at various points. ●

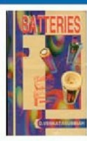
The author is a B.Tech in electronics & communication engineering

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